

Vidhyakshaya Kannan

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Education

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- Stanford University** Sep 2026 – June 2028
M.S. in Symbolic Systems
- Sai University** Sep 2022 – May 2026
B.Tech, Computing and Data Science; Minor in English Literature CGPA: 9.58/10
- Department rank 1
 - **Coursework:** Introduction to Programming, Data Structures and Algorithms, Data Analysis and Visualization, Multivariable Calculus, Linear Algebra, Probability and Statistics, Machine Learning, Deep Learning, Artificial Intelligence, Natural Language Processing, Computer Vision
- New England College** Aug 2024 – Dec 2024
Exchange Student, Computer Information Systems GPA: 4.0/4.0
- **One of five** students from India to be selected to participate in the Global Undergraduate Exchange Program, funded by the U.S. Department of State.
 - Awarded **full-ride scholarship worth USD 30,000** to pursue one semester of undergraduate study in Computer Science in the U.S.
 - Scholarship covered visa expenses, round trip airfare, tuition, required fees, on-campus housing, meals, health insurance and a monthly stipend.
 - **Coursework:** Object-Oriented Programming, Cybersecurity Foundations

Experience

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- Research Assistant** Atlanta, GA (Remote)
Georgia Tech Financial Services Innovation Lab; Supervised by Dr. Sudheer Chava May 2024 - Present
- **IPO-Mine: Structured Mining and Analysis of S-1 IPO Filings**
 - **Accepted to KDD 2026.**
 - Developed *IPO-Mine*, a pip-installable, scalable tool to parse, normalize, and segment S-1 and F-1 IPO filings, enabling systematic analysis of firm disclosures across sections and time periods.
 - Evaluated parser using large-scale dataset of S-1 filings from 1995-2025, achieving an overall extraction accuracy of 90%+.
 - Implemented LLM-based validation of parsed sections to improve accuracy of the tool.
 - Conducted studies correlating extracted text to downstream financial and market outcomes, and contributed to evaluation of multimodal LLMs.
 - **ConfReady: A RAG based Assistant and Dataset for Conference Checklist Responses** 📄 | [Paper](#)
 - Developed *ConfReady*, a [pip-installable](#) RAG based tool to assist authors with ARR Responsible NLP Research checklist requirements.
 - Built contextual RAG backend and performed quantitative analysis on a stratified sample of research papers with checklists, improving accuracy of the tool.
 - **Co-first author paper accepted to EMNLP 2025 (System Demonstrations Track).**
 - **KG-MuLQA: A Knowledge-Graph-Based Framework for Systematic Question Generation and Long-Context LLM Evaluation** 📄 | [Paper](#)
 - **Accepted to ACL 2026 (Oral).**
 - Designed largest publicly available benchmark (20K+ questions) to evaluate long-context language models' ability to retrieve and process information from financial documents.
 - Extracted questions at multiple complexity levels using structured knowledge-graph based representations of financial agreements.

- Evaluated 11 SoTA LLMs and reported decline in performance with increasing question complexity. Conducted detailed error analysis, ablations and identified various failure modes.

Data Science Intern

The Gen Academy

Bangalore, KA (Remote)

Oct 2025 - Present

- Conducted research on context engineering, fine-tuning strategies, and multi-agent systems.
- Developed course materials, labs and exercises for artificial intelligence workshops attended by 250+ mid to senior level industry professionals.
- Developed n8n workflows to automate promotional content generation for social media and emails.

Undergraduate Student Researcher

Sai University; Supervised by Dr. Jamshed Bharucha, Dartmouth College

Chennai, TN

Aug 2023 - Present

- Designed experiments in Python to investigate auditory temporal attention and its role in enhancing music perception through the synchronization of neural oscillators with rhythmic sequences.
- Collected responses from over 15 participants and performed detailed statistical analysis.

Research Intern

NetoAI

Chennai, TN

May 2025 - Jul 2025

- Developed T-VEC, the telecommunications industry's first domain-specific embedding model (**20K+ downloads on HuggingFace**).
- Fine tuned **gte-Qwen2-1.5B-instruct** using deep triplet loss on 100K domain-specific triplets.
- Co-first author paper accepted to EMNLP 2025 (Industry Track). Preprint: [arXiv:2504.16460](https://arxiv.org/abs/2504.16460)
- Developed an end to end low latency voice to voice agent with streaming ASR, real time TTS and quantized LLMs and achieved end to end latency of under 0.5s.

Software Developer Intern

NetoAI

Chennai, TN (Remote)

June 2024 – Sep 2024

- Designed and implemented a network inventory reconciliation application in Spring Boot, enabling alignment between the real-world telecom network state and its digital twin representation.
- Optimized comparison algorithm, reducing reconciliation time by 70% for large-scale network topologies.

Software Engineer Intern

Prodapt

Chennai, TN

Aug 2022 – Sep 2023

- Contributed to migration of NetBots platform from monolithic to microservices, decomposing services, containerizing with Docker, and orchestrating with Kubernetes.
- Built SpringBoot APIs for [NetInsight](#), network life cycle inventory management tool, used in production by telecom clients.

Publications

Galarnyk, M., Lohani, S., **Kannan, V.**, et al. **IPO-Mine: A Toolkit and Dataset for Section-Structured Analysis of Long, Multimodal IPO Documents**. In Proceedings of the 32nd ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD '26), pages 8882–8893. ACM, 2026.

Galarnyk, M.*, Routu, R.*, **Kannan, V.***, Bheda, K., Banerjee, P., Shah, A., and Chava, S. **ConfReady: A RAG based Assistant and Dataset for Conference Checklist Responses**. In Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing: System Demonstrations, pages 667–677, Suzhou, China. Association for Computational Linguistics.

Tatarinov, N., **Kannan, V.***, Srinivasa, H.*, Raj, A., Anand, H. S., Singh, V., Luthra, A., Lade, R., Shah, A., and Chava, S (2025). “**KG-MuLQA: A Knowledge-Graph-Based Framework for Systematic Question Generation and Long-Context LLM Evaluation**.” Accepted to ACL 2026 (Oral). Preprint: [arXiv:2505.12495](https://arxiv.org/abs/2505.12495)

Ethiraj, V.*, David, A.*, Menon, S.*, Vijay, D.*, and **Kannan, V.***. **T-VEC: A Telecom-Specific Vectorization Model with Enhanced Semantic Understanding via Deep Triplet Loss Fine-Tuning**. In Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing: Industry Track, pages 2449–2460, Suzhou (China). Association for Computational Linguistics.

Awards and Honors

Global UGRAD Scholar Bureau of Educational and Cultural Affairs, U.S Department of State
Invited Member, Young South Asian Leaders Initiative U.S Department of State
Emerging Technologist Millennium Fellow United Nations Academic Impact

Certifications

Generative AI with Large Language Models *DeepLearning.AI (Coursera)*
Deep Learning *Indian Institute of Technology Madras (NPTEL)* (**top 5%** of 3779 students certified)
Understanding and Applying Text Embeddings *DeepLearning.AI*
Embedding Models: From Architecture to Implementation *DeepLearning.AI*
Introduction to Discrete Mathematics for Computer Science *University of California, San Diego (Coursera)*
Computer Networks and Internet Protocol *Indian Institute of Technology Kharagpur (NPTEL)* (**top 5%** of 3082 students certified)

Technical Skills

Languages: Python, C, Java, Julia; **Database:** MySQL, PostgreSQL, Neo4j; **Machine Learning/NLP:** Pandas, NumPy, PyTorch, TensorFlow, scikit-learn, Hugging Face Transformers, LangChain, LlamaIndex, spaCy, NLTK, tokenizers; **Tools/Technologies:** Kubernetes, Docker, Apache Kafka, Git/GitHub; **Frameworks:** Django, FastAPI, Flask, SpringBoot

Positions of Responsibility

Peer Tutor Served as peer tutor for *Introduction to Programming, Calculus, Data Structures and Algorithms*, and *Introduction to Embedded Systems and Robotics* at Sai University. Mentored 120+ students across four semesters. Designed problem sets and held weekly office hours for course assistance.

President, Computer Science Society Led a 7-member core team to conduct 10+ events, technical workshops, faculty lectures, and coding competitions with 300+ cumulative participants. Partnered with 4+ industry professionals to deliver career talks and hands-on tutorials. Oversaw creation of mentorship programs for first-year CS students.

Secretary of Student Academics, Undergraduate Student Government Elected by the student body to represent 250+ undergraduates. Coordinated with deans and academic departments to optimize timetables for 40+ courses each semester. Organized 6+ academic and professional development events in collaboration with faculty.

Member, Anti-Ragging Committee Drafted and implemented Sai University's anti-ragging policy. Designed a confidential online reporting system used by 300+ students. Conducted sensitization workshops for incoming cohorts and coordinated with administration to ensure compliance with UGC guidelines.

Additional Activities

Reviewer, 19th Conference of the European Chapter of the Association for Computational Linguistics

Panelist, *Computing and Data Science: Societies, Startup and the Future*. One of three students to participate in panel discussion held at Sai University with Prof. John C. Mitchell, Mary and Gordon Crary Family Professor, Professor of Computer Science, **Stanford University**.

Attendee, *NeuroAI 2024: Center for Cognitive Neuroscience Workshop on Neuroscience and Artificial Intelligence*. Attended sessions on neural computations, brain-wide population dynamics, machine learning models of human cognition, and explainable neural networks at **Dartmouth College**.

Speaker, *Containerization in 30 Minutes: From Python to Kubernetes*. Delivered a technical talk on containerizing Python applications and deploying them with Kubernetes.

* Indicates equal contribution.